

Health Economics – Assignment 5

Part I – MC (2 points each)

Question 1. Network and referrals

B. She can see the in-network dermatologist without a referral, paying the usual in-network cost-sharing.

Question 2. HDHP and cost-sharing

B. \$2,200

Calculation: She pays the first \$1,500 (deductible). The remaining expense is \$3,500. She pays 20% of \$3,500, which is \$700. Total out-of-pocket is \$1,500 + \$700 = \$2,200.

Question 3. Ex ante vs ex post moral hazard

B. A person with full coverage for physical therapy schedules extra sessions mainly for mild soreness.

Question 4. Expected utility and buying insurance

C. The premium is higher than the expected loss, but they are willing to pay extra to avoid financial risk.

Question 5. Moral hazard in physical therapy use

D. Quantity demanded falls, and patient spending per session rises.

Bonus Question: With full insurance, the marginal out-of-pocket cost to the patient is \$0, leading them to consume until their marginal benefit is zero. When uninsured, the patient faces the full market price (\$80), causing them to reduce consumption, while their per-session spending naturally rises from \$0 to the market price of \$80.

Part II – Problems

Problem 1. Adverse Selection in an insurance market (10 points)

1a) Average expected cost and actuarially fair premium (2 points)

$$\begin{aligned}\text{Total Expected Cost} &= 250 \times \$400 + 250 \times \$1,200 = \$400,000 \\ \text{Average Expected Cost} &= \frac{\$400,000}{500} = \$800\end{aligned}$$

The actuarially fair community premium for the plan would be \$800.

1b) Willingness to enroll at the average cost premium (3 points)

- **Low-risk individuals:** No. The premium (\$800) is higher than their expected costs (\$400), so they will not enroll.
- **High-risk individuals:** Yes. The premium (\$800) is lower than their expected costs (\$1,200), making it financially beneficial for them to enroll.

1c) New average expected cost and premium adjustment (2 points)

If low-risk individuals drop out, the new average expected cost among enrollees (only high-risk individuals) is **\$1,200**. If the insurer adjusts the premium to equal this new average cost, the premium will **rise** from \$800 to \$1,200.

1d) Adverse selection and the death spiral (3 points)

This example illustrates adverse selection because the community-rated premium disproportionately attracts sicker, high-cost individuals, while driving away healthier individuals. As healthy people drop out, the average cost of the remaining pool increases, forcing the insurer to raise premiums further; this creates a feedback loop where healthy people keep dropping out as premiums rise, eventually leading to a “death spiral” where the market may collapse.

Problem 2. Expected loss, fair premium, and risk aversion (10 points)**2a) Expected medical spending (2 points)**

$$\text{Expected Spending} = 0.02 \times \$50,000 + 0.98 \times \$0 = \$1,000$$

2b) Actuarially fair premium (2 points)

The actuarially fair premium is exactly equal to the expected loss, which is **\$1,000**.

2c) Maximum administrative load (3 points)

The maximum dollar administrative load is **\$500**. The consumer will purchase the policy as long as the total premium does not exceed their maximum willingness to pay (\$1,500). Since the expected loss (fair premium) is \$1,000, the insurer can add up to \$500 in administrative load (\$1,500 - \$1,000 = \$500).

2d) Relationship between expected loss, administrative load, and willingness to pay (3 points)

An individual's willingness to pay for insurance is the sum of their expected loss and the risk premium they are willing to pay to eliminate financial uncertainty. Because this person is risk-averse, their willingness to pay (\$1,500) exceeds their expected loss (\$1,000), which creates a margin that justifies the insurer charging an above-fair premium to cover the administrative load (up to \$500) while still making the transaction mutually beneficial.